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Probabilistic Artificial Intelligence
Questions Pack

January 19, 2026

Time limit: 120 minutes

Instructions. This pack contains all questions for the final exam. It contains the questions only. You must provide your answers to the questions on the accompanying **answer sheet** by blackening out the corresponding squares. Please carefully read the following instructions regarding the answer sheet:

- During the exam, you may use pencil and eraser to mark and edit your answers on the answer sheet. After the time is up, we will collect the questions pack. Make sure that you marked all your answers by then, at least with a pencil. After collecting the questions pack, we provide you with additional **10** minutes of time to blacken out the squares on the answer sheet with a **black pen**.
- As the answer sheet will be graded by a computer, please **make sure to blacken out the whole square. Do not use ticks or crosses**.
- Nothing written on the pages of the question pack will be collected or marked. **Only the separate answer sheet with the filled squares will be marked**.
- Please make sure that your answer sheet is clean and do not write anything on the answer sheet except the squares you blackened out. We reserve the right to classify answers as wrong without further consideration if the sheet is filled out ambiguously.

Collaboration on the exam is strictly forbidden. You are allowed a summary of *two* A4 pages and a simple, non-programmable calculator. The use of any other helping material or collaboration will lead to being excluded from the exam and subjected to disciplinary measures by the ETH Zurich disciplinary committee.

Question Types In this exam, questions award **1, 2, 3, or 4** points if answered correctly, depending on the difficulty of the question. You will encounter the following question types:

- **True or False.**
- **Multiple choice questions with a *single* correct answer.**

  These multiple choice questions have **exactly one** correct choice.
- **Multiple choice questions with *multiple* correct answers.**

  These multiple choice questions are marked with a "♣" and have **one or more** correct choices (at least one choice is correct). You will receive full points if you have chosen **all** the correct choices and **none** of the wrong choices. You will get zero points otherwise.

There are no negative points in this exam, so **0 points** are awarded if a question is answered wrongly or not attempted. The sections and questions are not ordered by difficulty. Thus, if you find a question too difficult, it may make sense to get back to it in the end.

Questions in the same section might be related. In addition, some questions require information from previous parts. Hence, if you intend to skip some questions, make sure that you read all notes and details between the questions.

The notation $\mathcal{N}(\mu, \sigma^2)$ denotes the Gaussian random distribution with mean μ and variance σ^2. For example, $\mathcal{N}(3, 4)$ designates the Gaussian distribution with mean 3 and standard deviation 2. $\log(\cdot)$ denotes the natural logarithm.

Good luck!

1 Bayesian Linear Regression (9 points)

Question 1 *(1 point)*
Bayesian Linear Regression does not allow to separate epistemic from aleatoric uncertainty.

A True B False

Question 2 *(1 point)*
Assuming noise is (cond.) i.i.d. Gaussian, and the prior on the model parameters is Gaussian, ridge regression can be understood as finding the Maximum Likelihood (MLE) parameter estimate for a linear regression problem.

A True B False

Question 3 *(2 points)*
We consider a generalized Bayesian Linear Regression. We are given a dataset of inputs $\mathbf{X} \in \mathbb{R}^{n \times d}$, and $\mathbf{y} \in \mathbb{R}^n$. Consider the model

$$\mathbf{y} = \mathbf{X}\mathbf{w} + \epsilon$$

Unlike the standard setting, we assume that the noise is correlated with a covariance matrix $\mathbf{\Sigma}_n \in \mathbb{R}^{n \times n}$, $\epsilon \sim \mathcal{N}(0, \mathbf{\Sigma}_n)$. We place a prior on the weights with a non-zero mean $\mathbf{m}_p \in \mathbb{R}^d$ and a covariance matrix $\mathbf{\Sigma}_p \in \mathbb{R}^{d \times d}$, that is, $\mathbf{w} \sim \mathcal{N}(\mathbf{m}_p, \mathbf{\Sigma}_p)$. What is the marginal distribution of $\mathbf{y}$ given inputs $\mathbf{X}$?

A $p(\mathbf{y} \mid \mathbf{X}) = \mathcal{N}(\mathbf{X}\mathbf{m}_p, \mathbf{X}\mathbf{\Sigma}_p\mathbf{X}^T + \mathbf{\Sigma}_n)$

B $p(\mathbf{y} \mid \mathbf{X}) = \mathcal{N}(\mathbf{0}, \mathbf{X}\mathbf{\Sigma}_p\mathbf{X}^T + \mathbf{\Sigma}_n)$

C $p(\mathbf{y} \mid \mathbf{X}) = \mathcal{N}(\mathbf{X}\mathbf{m}_p, (\mathbf{\Sigma}_p^{-1} + \mathbf{X}^T\mathbf{\Sigma}_\mathbf{n}^{-\mathbf{1}}\mathbf{X})^{-1})$

D $p(\mathbf{y} \mid \mathbf{X}) = \mathcal{N}(\mathbf{X}\mathbf{m}_p, \mathbf{\Sigma}_p + \mathbf{\Sigma}_n)$

Question 4 *(2 points)*
Assume that the prior over the bias for flipping a coin is distributed as $\text{Beta}(\theta; \alpha, \beta)$. We model the likelihood of the outcome from the coin flips with a binomial distribution as $k \mid \theta \sim \text{Bin}(n, \theta)$ where θ is the probability of heads. Let $n_H = k$ be the number of heads, and $n_T = n - k$ the number of tails in the binomial trial. Then the posterior distribution of the coin given the binomial trials is distributed as $\theta \mid k \sim \text{Beta}(\alpha + n_T, \beta + n_H)$.

Hint: $\text{Bin}(k; n, \theta) = \binom{n}{k}\theta^k(1-\theta)^{n-k}$, and $\text{Beta}(\theta; \alpha, \beta) \propto \theta^{\alpha-1}(1-\theta)^{\beta-1}$.

A True B False

Question 5 *(1 point)*
Bayesian linear regression can be implemented online where the data $(\mathbf{x} \in \mathbb{R}^d, y \in \mathbb{R})$ is coming one-by-one in an i.i.d. stream, and the posterior distribution is updated recursively. The algorithm can be implemented with $\mathcal{O}(d)$ memory.

A True B False

Question 6 *(2 points)*
Let $(X, Y) \sim \mathcal{N}(0, \Sigma)$, where $\Sigma = \begin{pmatrix} 1 & \alpha \\ \alpha & 1 \end{pmatrix}$, and $\alpha \in \mathbb{R}, |a| \leq 1$. Then random vectors $U = X + Y$ and $V = X - Y$ are independent if and only if $\alpha = 0$.

A True B False

2 Gaussian Processes (13 points)

2.1 Kernels (7 points)

Question 7 *(2 points)* Let $k : \mathbb{R} \times \mathbb{R} \to \mathbb{R}$ be a valid kernel, then

$$g(x, x') \triangleq \int_{-\infty}^{\infty} k(x + u, x' + u) \exp\left(-u^2\right) \mathrm{d}u$$

is a valid kernel. You may assume that the integral exists.

A True B False

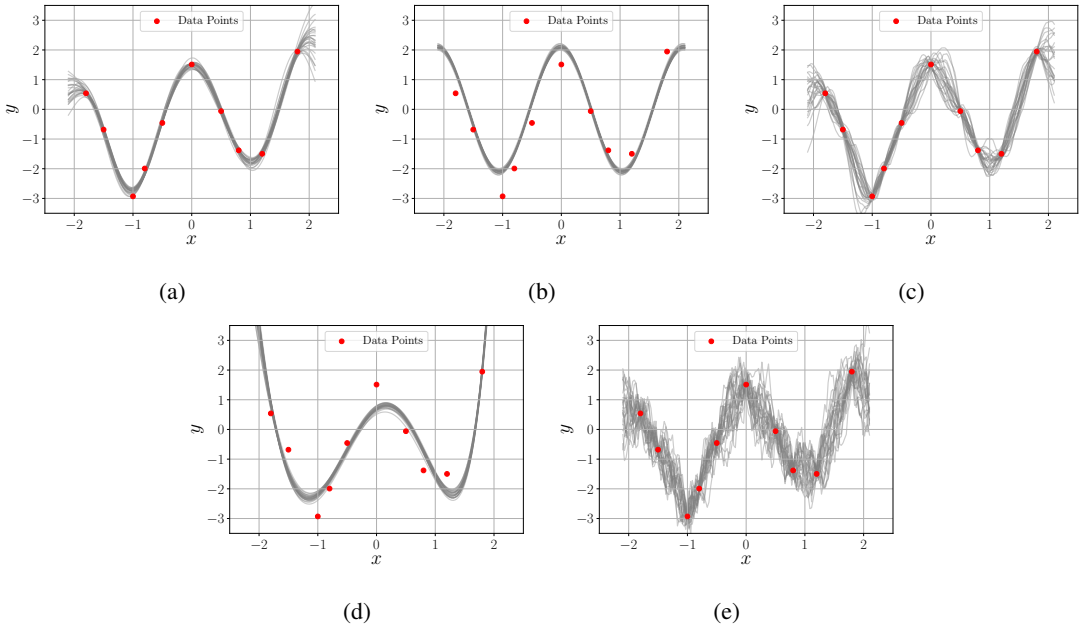


Figure 1: Samples of posterior Gaussian processes for different kernels k. The red dots represent the data $\mathcal{D}$.

Suppose we are given a dataset $\mathcal{D} = \{(x_1, y_1), ..., (x_n, y_n)\}$ of n observations. We fit Gaussian processes with zero mean and the following five kernels to the data $\mathcal{D}$.

- $k_1(x, x') = (1 + xx')^5$
- $k_2(x, x') = \cos\left(3\,|x - x'|\right)$
- $k_3(x, x') = \exp\left(-2\,(x - x')^2\right)$
- $k_4(x, x') = \exp\left(-2\,|x - x'|\right)$
- $k_5(x, x') = \left(1 + 2\sqrt{3}\,|x - x'|\right) \cdot \exp\left(-2\sqrt{3}\,|x - x'|\right)$

Representative samples from posterior Gaussian processes are depicted in Fig. 1. Match each of the plots in Fig. 1 to the corresponding kernel. There is a one-to-one correspondence between the kernel functions $\{k_1, ..., k_5\}$ and the plots in Fig. 1.

Question 8 *(1 point)* Which plot corresponds to the kernel k_1?

A Plot (a) B Plot (b) C Plot (c) D Plot (d) E Plot (e)

Question 9 *(1 point)* Which plot corresponds to the kernel k_2?

A Plot (a) B Plot (b) C Plot (c) D Plot (d) E Plot (e)

Question 10 *(1 point)* Which plot corresponds to the kernel k_3?

A Plot (a) B Plot (b) C Plot (c) D Plot (d) E Plot (e)

Question 11 *(1 point)* Which plot corresponds to the kernel k_4?

A Plot (a) B Plot (b) C Plot (c) D Plot (d) E Plot (e)

Question 12 *(1 point)* Which plot corresponds to the kernel k_5?

A Plot (a) B Plot (b) C Plot (c) D Plot (d) E Plot (e)

2.2 Bayesian Model Selection (6 Points)

Consider a regression model where the target $y \in \mathbb{R}$ is generated from inputs $x \in \mathbb{R}$ according to:

$$y = f(x) + \varepsilon, \qquad \varepsilon \sim \mathcal{N}(0, \sigma^2),$$

for an unknown function f. Assume the noise variance is known to be $\sigma^2 = 1$. We place a Gaussian Process prior on the function f, denoted as $f \sim \mathrm{GP}(0, k)$, using a cosine kernel with an angular frequency parameter ω:

$$k(x, x') = \cos\big(\omega\,|x - x'|\big).$$

We observe a dataset $\mathcal{D}$ consisting of two points:

$$(x_1, y_1) = (0, 1) \qquad (x_2, y_2) = (\pi/2, 0).$$

To tune the angular frequency ω of the kernel, we maximize the marginal likelihood $p(y_1, y_2 \mid x_1, x_2, \omega)$.

Question 13 *(4 points)* Which of the following sets contains *all* values of ω that represent stationary points (critical points) of the log-likelihood function $\log p(y_1, y_2 \mid x_1, x_2, \omega)$?

Hint I: The marginal log-likelihood is given by

$$\log p(y_1, y_2 \mid x_1, x_2, \omega) = -\frac{1}{2}\,\boldsymbol{y}^T \big(\mathbf{K} + \sigma^2 \mathbf{I}\big)^{-1} \boldsymbol{y} - \frac{1}{2} \log\det(\mathbf{K} + \sigma^2 \mathbf{I}) + \text{const.}$$

Hint II: The determinant of a 2×2 *matrix is given by*

$$\det \begin{bmatrix} a & b \\ c & d \end{bmatrix} = ad - bc.$$

Hint III: The inverse of an invertible 2×2 *matrix is given by*

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}.$$



A $\{\omega = 2k \text{ for } k \in \mathbb{Z}\}$

B $\{\omega = 2k \text{ for } k \in \mathbb{N}_{\geq 0}\}$

C $\{\omega = 2k + 1 \text{ for } k \in \mathbb{Z}\}$

D $\{\omega = k \text{ for } k \in \mathbb{Z}\}$

E $\{\omega = \pi k \text{ for } k \in \mathbb{Z}\}$

In the previous question, we established that the log-likelihood $\log p(y_1, y_2 \mid x_1, x_2, \omega)$ has a countably infinite set of critical points. Upon closer inspection, we see that these points alternate between local maxima and minima, leaving us with an infinite number of Maximum Likelihood Estimators (MLEs). To resolve this non-uniqueness, we place a hyperprior on ω. This acts as a form of regularization, allowing us to single out a unique MAP estimate.

Question 14 *(2 points)* Let us impose a hyperprior on ω with the following probability distribution:

$$\mathbb{P}(\omega = k) = \begin{cases} \frac{1}{2^{k+1}} & k \in \mathbb{N}_{\geq 0} \\ 0 & \text{otherwise.} \end{cases}$$

Calculate the MAP estimate $\hat{\omega}$, defined as the solution to the optimization problem:

$$\hat{\omega} = \arg\max_{\omega} \log p(\omega \mid x_1, x_2, y_1, y_2).$$

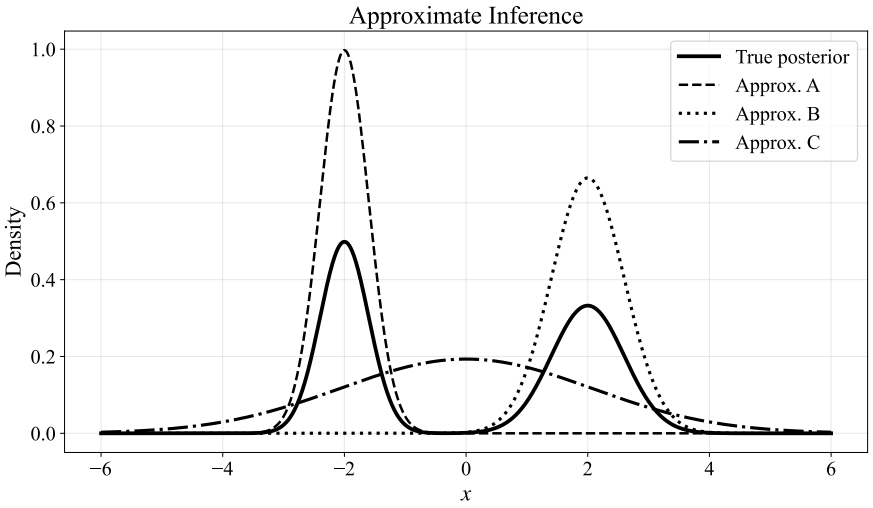
Hint: You may use the approximations:

- $\log 2 \approx 0.69$
- $\log 3 \approx 1.10$
- $\log 4 \approx 1.39$

A $\hat{\omega} = 1$ B $\hat{\omega} = 0$ C $\hat{\omega} = 2$ D $\hat{\omega} = 3$ E $\hat{\omega} = -1$

3 Variational Inference & Diffusion (12 points)

3.1 Variational Inference (7 points)



In the plot above, the solid black curve is a bimodal posterior distribution $p(x \mid y)$. The three other curves (A, B, C) are Gaussian approximations produced by three different approximate inference methods. Match each curve to the corresponding method:

- Laplace approximation;
- Variational inference with a Gaussian $q(x) = \mathcal{N}(\mu, \sigma^2)$ minimizing reverse-KL, $\mathrm{KL}(q\|p)$;
- Variational inference with a Gaussian $q(x) = \mathcal{N}(\mu, \sigma^2)$ minimizing forward-KL, $\mathrm{KL}(p\|q)$.

Question 15 *(1 point)* Which method corresponds to approximation A?

A Laplace B Reverse-KL C Forward-KL

Question 16 *(1 point)* Which method corresponds to approximation B?

A Laplace B Reverse-KL C Forward-KL

Question 17 *(1 point)* Which method corresponds to approximation C?

A Laplace B Reverse-KL C Forward-KL

Question 18 ♣ *(4 points)* Consider a 1D Bayesian model

$$y = w + \epsilon, \qquad \epsilon \sim \mathcal{N}(0, \sigma_n^2),$$

with prior $w \sim \mathcal{N}(0, \sigma_p^2)$. We approximate the posterior $p(w \mid y)$ by a Gaussian variational distribution $q(w) = \mathcal{N}(m, s^2)$. The ELBO is

$$\mathcal{L}(m, s^2) = \mathbb{E}_q[\log p(y \mid w)] - \mathrm{KL}(q \| p(w)).$$

Let m_{opt} and s^2_{opt} be the maximizers of $\mathcal{L}$.
Hint: A 1D Gaussian $x \sim \mathcal{N}(\mu, \sigma^2)$ has the following probability density function

$$p(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{1}{2}\frac{(x-\mu)^2}{\sigma^2}\right).$$

The variance of a random variable X can be calculated by $Var(X) = \mathbb{E}(X^2) - \mathbb{E}(X)^2$.
The KL divergence between two 1D Gaussians $p_0 = \mathcal{N}(\mu_0, \sigma_0^2)$ and $p_1 = \mathcal{N}(\mu_1, \sigma_1^2)$ is

$$\mathrm{KL}(p_0 \| p_1) = \log\frac{\sigma_1}{\sigma_0} + \frac{\sigma_0^2 + (\mu_0 - \mu_1)^2}{2\sigma_1^2} - \frac{1}{2}.$$

Which of the following statements are **true**? **Mark all that apply.**

A As $\sigma_n^2 \to 0$, $s^2_{\text{opt}} \to 0$.

B As $\sigma_n^2 \to \infty$, $m_{\text{opt}} \to 0$.

C As $\sigma_n^2 \to \infty$, $s^2_{\text{opt}} \to 0$.

D Increasing σ_p^2 always decreases s^2_{opt}.

3.2 Diffusion (5 Points)

Question 19 *(1 point)* Diffusion is a form of supervised learning that requires labeled data $\{x^{(i)}, y^{(i)}\}_{i=1}^n$.

A True B False

Question 20 *(1 points)*
A longer diffusion process in the DDPM model, i.e., larger T, makes sampling new datapoints computationally more expensive.

A True B False

Question 21 *(1 points)*
The training time per SGD iteration increases with larger time-horizon T in the DDPM training procedure as covered in the lecture.

A True B False

Question 22 *(2 points)*
Consider a diffusion on the domain $\mathbb{R}$ with the following one-step forward-process distribution

$$q(x_t|x_{t-1}) = \mathcal{N}(x_t; \sqrt{1-\beta_t}x_{t-1}, 2\beta_t)$$

where $\beta_t \in (0, 1]$ is a monotonically increasing sequence. What is the marginal distribution $q(x_t|x_0)$ in the limit when $t \to \infty$?
Hint: The distribution of the convolution of gaussian random variables can be calculated as follows.

$$\mathcal{N}(y; a\mu, b + a^2\sigma^2) = \int \mathcal{N}(x; \mu, \sigma^2)\mathcal{N}(y; ax, b)dx$$

where $\mathcal{N}(x; \mu, \sigma^2)$ and $\mathcal{N}(y; ax, b)$.

A $q(x_t|x_0) = \mathcal{N}(2, 1)$

B $q(x_t|x_0) = \mathcal{N}(0, 1)$

C $q(x_t|x_0) = \mathcal{N}(0, 2)$

D $q(x_t|x_0) = \mathcal{N}(0, 4)$

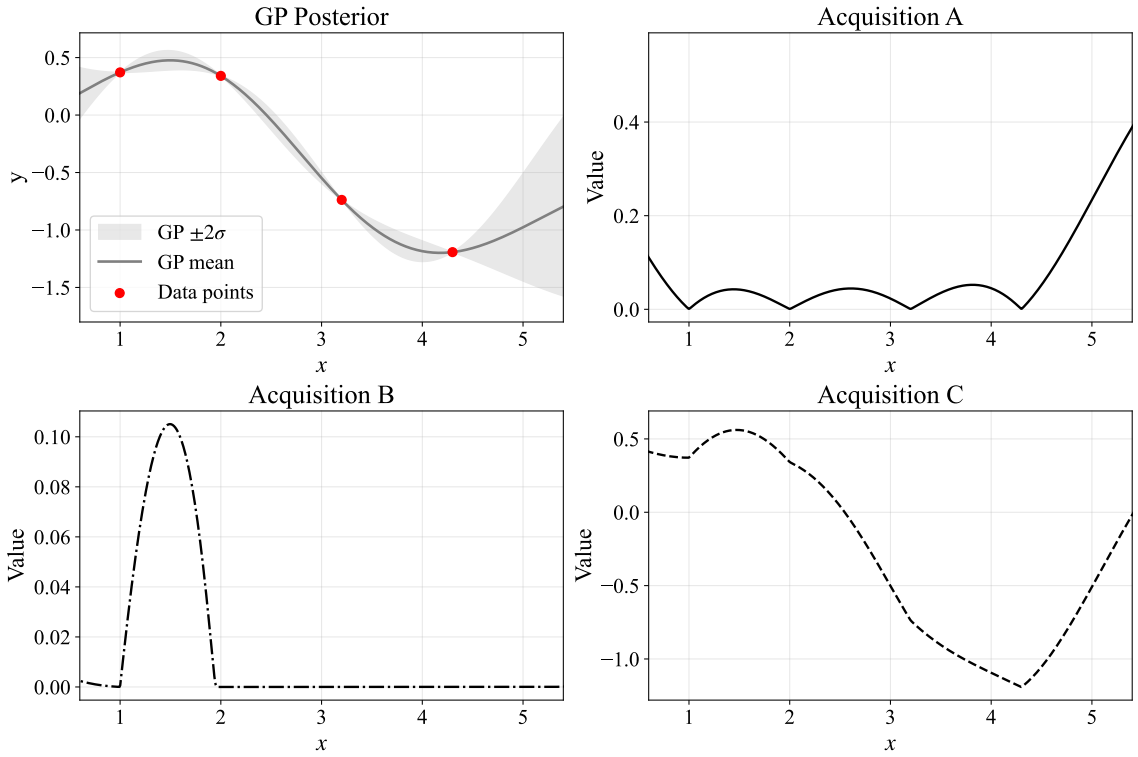
4 Active Learning & Bayesian Optimization (8 points)

Question 23 *(1 point)* In a Gaussian Process $p(f) = \text{GP}(f; \mu, k)$ with Gaussian observation noise $y = f(x) + \epsilon, \epsilon \sim \mathcal{N}(0, \sigma_n^2)$, the mutual information $I(f(x); y)$ increases as the noise variance σ_n^2 increases.

A True B False

Question 24 *(1 point)* Thompson Sampling with a Gaussian process model selects a point x by:

A Maximizing the posterior mean.
B Minimizing posterior variance.
C Sampling a function from the Gaussian process posterior and choosing the maximizer of that sample.
D Selecting the point with highest expected improvement.



Consider the figure above showing a Gaussian process posterior together with three acquisition functions labeled A, B, and C. Let $\mu(x)$ and $\sigma(x)$ denote the GP predictive mean and standard deviation, and let f_{best} be the best observed function value so far.
The following acquisition functions are defined as:

- Uncertainty sampling (US):
$$a_{\text{US}}(x) = \sigma(x)$$
- Upper confidence bound (UCB) with $\beta > 0$:
$$a_{\text{UCB}}(x) = \mu(x) + \beta\sigma(x)$$

- Expected improvement (EI):

$$a_{\text{EI}}(x) = \mathbb{E}[\max(0, f(x) - f_{\text{best}})]$$

Match each acquisition function to the corresponding curve in the figure.

Question 25 *(1 point)* Which acquisition function corresponds to curve A?

A US B UCB ($\beta = 2$) C EI

Question 26 *(1 point)* Which acquisition function corresponds to curve B?

A US B UCB ($\beta = 2$) C EI

Question 27 *(1 point)* Which acquisition function corresponds to curve C?

A US B UCB ($\beta = 2$) C EI

Question 28 ♣ *(3 points)* Consider a Bayesian binary classification with latent parameter θ. For an input x, denote

$$p_\theta = p(y = 1 \mid x, \theta).$$

Assume an approximate posterior $q(\theta)$, and define the BALD (Bayesian Active Learning by Disagreement) acquisition score

$$I(\theta; y_x) = H(\mathbb{E}_{\theta\sim q}[p_\theta]) - \mathbb{E}_{\theta\sim q}[H(p_\theta)],$$

where $H(p) = -p \log p - (1 - p) \log(1 - p)$ is the Bernoulli entropy.
Which of the following statements are **true**? **Mark all that apply.**

A If all posterior samples predict the same probability $p_\theta = p_0$, then the BALD score at x is zero.

B The BALD score depends only on the predictive mean $\mathbb{E}_\theta[p_\theta]$, not on its variance.

C Consider two cases: (A) uncertainty is aleatoric (i.e., $p_\theta = 0.5$); and (B) uncertainty is epistemic (i.e., $p_\theta = 0$ and $p_\theta = 1$ with probability 0.5 each). The BALD scores for cases (A) and (B) are different.

5 MDPs & Reinforcement Learning (18 points)

5.1 Influence of the action cost (6 points)

For the next two questions, consider an MDP with an infinite set of states $S = \{1, 2, 3, \dots\}$ and start state $s = 1$. Each state allows a continuous action $a \in [0, 1]$. For every state s, the transitions are:

$$P(s+1 \mid s, a) = a, \qquad P(s \mid s, a) = 1 - a.$$

The rewards are defined as follows:

$$r(s, a, s+1) = 1 - \lambda a^2, \qquad r(s, a, s) = -\lambda a^2,$$

where $\lambda \geq 0$ represents action cost regularization. The discount factor is $\gamma = 0.5$.

Question 29 *(2 points)* First, assume that there is no action regularization, i.e., $\lambda = 0$. What is the optimal policy $\pi(s)$ in this case?

A $\pi(s) = \frac{1}{s^2}$ B $\pi(s) = \frac{1}{s}$ C $\pi(s) = 1$ D $\pi(s) = \frac{1}{2}$

Question 30 *(4 points)* Now, assume that we penalize action with $\lambda = 2$. What is the optimal value $V^*(1)$ in this case?

A $V^*(1) = \frac{1}{4}$ B $V^*(1) = \frac{1}{2}$ C $V^*(1) = 1$ D $V^*(1) = 2$

5.2 True/False questions (5 points)

For the next four questions, assume a finite MDP with discount factor $\gamma \in (0, 1)$.

Question 31 *(1 point)* If two deterministic policies π_1, π_2, satisfy $V^{\pi_1}(s) = V^{\pi_2}(s)$ for every state s, then π_1 and π_2 must be identical.

A True B False

Question 32 *(1 point)* For any value function V, the greedy policy

$$\pi(x) = \arg\max_a \Big[r(x, a) + \gamma \sum_{x'} P(x' \mid x, a)\, V(x') \Big]$$

always satisfies $V^\pi = V$.

A True B False

Question 33 *(2 points)* Let W, V be two value functions such that

$$W(x) \leq V(x) \quad \text{for all } x.$$

Let B denote the Bellman operator, defined by

$$(BU)(x) \coloneqq \max_a \Big[r(x, a) + \gamma \sum_{x'} P(x' \mid x, a)\, U(x') \Big]$$

for any value function U, where $0 \leq \gamma < 1$. Then it holds that

$$(BW)(x) \leq (BV)(x) \quad \text{for all } x \in \mathcal{X}.$$

A True B False

Question 34 *(1 point)* Policy iteration converges to an optimal policy π^* in finite iterations.

A True B False

5.3 Applying TD & Q-learning (3 points)

Consider an observed transition (x, a, r, x', a'), with $r = 4.5$. The current estimates are $V(x) = 5$, $Q(x, a) = 2$, and $V(x') = 7$. Use discount $\gamma = 0.5$ and learning rate $\alpha = 0.25$.

Question 35 *(2 points)* The TD(0) update produces the new value:
Hint: TD(λ) learns from a weighted average of n-step returns. For TD(0), the update uses only the one-step return:

$$V(x) \leftarrow V(x) + \alpha\Big(r + \gamma V(x') - V(x)\Big).$$

A $V(x) = 5$ B $V(x) = 5.75$ C $V(x) = 6.5$ D $V(x) = 7$

Question 36 *(1 point)* Q-learning is off-policy because its update target uses

$$r + \gamma \max_{a'} Q(x', a')$$

even when the action taken in x' was chosen by another policy.

A True B False

5.4 Policy Gradient Theorem for LQR (4 points)

Consider a discounted infinite-horizon Linear Quadratic Regulator (LQR) system

$$x_{t+1} = Ax_t + Ba_t, \qquad r(x_t, a_t) = -x_t^\top C x_t - a_t^\top D a_t,$$

with discount factor $0 < \gamma < 1$, where the reward is the negative quadratic cost.
We use a linear-Gaussian policy

$$a_t \sim \mathcal{N}(Kx_t,\ \Sigma),$$

where K is the trainable policy parameter and $\Sigma \succ 0$ is fixed. Let $J(K)$ denote the expected infinite-horizon discounted reward, and let d_K denote the (normalized) discounted state distribution under the policy parameter K and $\pi_K(a_t|x_t)$ the policy under K.

Question 37 *(1 point)* The policy gradient theorem implies

$$\nabla_K J(K) = \mathbb{E}_{x\sim d_K,\ a\sim\pi_K(\cdot|x)}\Big[\ (\nabla_K \log \pi_K(a|x))\ Q^{\pi_K}(x, a)\Big].$$

A True B False

Question 38 *(3 points)* For the linear-Gaussian policy

$$a = Kx + \varepsilon, \qquad \varepsilon \sim \mathcal{N}(0, \Sigma),$$

which expression for $\nabla_K \log \pi_K(a|x)$ is correct?



A $\nabla_K \log \pi_K(a|x) = (a - Kx)x^\top$

B $\nabla_K \log \pi_K(a|x) = -(a - Kx)x^\top$

C $\nabla_K \log \pi_K(a|x) = \Sigma^{-1}(a - Kx)x^\top$

D $\nabla_K \log \pi_K(a|x) = -\Sigma^{-1}(a - Kx)x^\top$